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B.E Semester-II (NEW) JUNE-2023

Subject: Fundamental of Mechanical Engineering

Date: 24/06/2023

Subject Code: CC-104-N

Time: 10:00 am to 01:00 pm

Total Marks: 70

Instructions:

1. Answer each section in separate Answer sheet.
2. Use of Scientific calculator is permitted.
3. All questions are **Compulsory**.
4. Indicate **clearly**, the options you attempt along with its respective question number.
5. Use the last page of main supplementary of **rough work**.

SECTION-I

- Que:1** (A) Write only the statement of Zeroth law, first law and second law of thermodynamics. [5]
- (B) What is prime mover? Also write down the various sources of energy. [5]
- (C) State Boyle's and Charle's laws. [5]

OR

- (C) Prove that work done during isothermal process is given by
 $W = p_1 v_1 \log(v_2/v_1) = p_1 v_1 \log(p_1/p_2)$. [5]

- Que:2**(A) Derive equation for Carnot vapour cycle. [5]
- (B) Write short note on separating calorimeter. [5]

OR

- (A) Derive thermal efficiency formula for Rankine cycle. [5]
- (B) Write short note on Dryness fraction and wet steam. [5]

- Que:3**(A) Explain with neat sketch the working of Babcock and Wilcox boiler [5]
- (B) Write short note on refrigerants used in refrigeration system. [5]

OR

- (A) What do you mean by mounting and accessories? Enlist list of boiler mountings and accessories. [5]
- (B) Explain with neat sketch Four stroke petrol engine. [5]

SECTION-II

- Que:4**(A) Difference between petrol and diesel engine. [5]
- (B) Define and explain Air compressor. [5]
- (C) Describe with neat sketch, the working of Watt Governor. [5]

OR

- (C) A 6 cylinder 4 stroke IC engine is to produce 95 KW brake power at 800 rpm. The stroke to bore ratio is 1.25. Mean effective pressure is 7 bar. Determine the bore and stroke of the engine. Assume mechanical efficiency as 80%. [5]

- Que:5**(A) What are the advantages and disadvantages of liquid fuels over solid fuels? [5]
- (B) Write a short note on: Types of belt drive. [5]

OR

- (A) Describe any three types of gears in short. [5]
- (B) Write a short on double acting reciprocating pump. [5]

- Que:6(A) Classify engineering materials & metals. [5]
(B) Explain working principle of steam turbines. [5]

OR

- (A) Define Welding, Soldering and Brazing. [5]
(B) 0.5 m^3 of air at pressure 1 bar is compressed at constant temperature until its volume is 0.2 m^3 . What is then its pressure? The air is then heated under constant pressure and its temperature is 75°C . Determine its volume and mass. [5]
Assume compression take place at 20°C and $R = 287 \text{ J/kg/K}$.

Best of luck